Volume I.

§ 37. The biquadratic equation.

A similar path to that used in solving the cubic equation by Cardano's formula may be taken for the solution of the equation of the fourth degree

$$y^4 + ay^2 + by + c = 0. \tag{1}$$

We put

$$2y = u + v + w$$

and obtain, if we write, for abbreviation,

$$s = u^2 + v^2 + w^2, \qquad t = v^2w^2 + w^2u^2 + u^2v^2, \tag{2}$$

then

$$4y^2 = s + 2(vw + wu + uv),$$
$$16y^4 = s^2 + 4s(vw + wu + uv) + 4t + 8uvw(u + v + w).$$

Substitution in (1) gives

$$s^2 + 4t + 4as + 16c + 8(uvw + b)(u + v + w) + 4(s + 2a)(vw + wu + uv) = 0.$$

This equation is satisfied by the assumptions

$$s + 2a = 0, \qquad uvw + b = 0, \qquad s^2 + 4t + 4as + 16c = 0.$$

With the help of the first of these three equations the third becomes

$$t = a^2 - 4c,$$

and, putting back the values (2) for s, t,

$$\begin{aligned} u^2 + v^2 + w^2 &= -2a, \\ v^2w^2 + w^2u^2 + u^2v^2 &= a^2 - 4c, \\ uvw &= -b. \end{aligned} \tag{3}$$

By § 7 these equations are satisfied if and only if u^2, v^2, w^2 are the roots of the cubic equation

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0, \tag{4}$$

and if the signs of u, v, w are so chosen that the last of equations (3) is satisfied. This last condition still allows four different determinations of signs, so that the four roots of the biquadratic equation are obtained in the following way:

$$\begin{aligned} 2\alpha &= u + v + w, \\ 2\beta &= u - v - w, \\ 2\gamma &= -u + v - w, \\ 2\delta &= -u - v + w. \end{aligned} \tag{5}$$

Thus the solution of the biquadratic equation has been reduced to that of the cubic equation (4).

This equation is called a cubic resolvent of the biquadratic equation.

§ 38. Proof of the fundamental theorem.

We now proceed to the proof of the fundamental theorem of algebra, namely that every equation $f(x) = 0$ has at least one root. We first put forward some general propositions.

1. If S denotes an arbitrary system of real numbers all greater than a certain positive or negative number C (that is, a system which contains no infinitely negative numbers), then there exists a lower bound for the numbers S.

By a lower bound g one is to understand a number which is not undercut by any number of the system S, but which is of such a nature that, if d is an arbitrarily given positive quantity, then between g and $g + d$ (including the endpoints) there always lies at least one number of the system S, however small d may be.

The proof follows directly from the possibility of a cut $(\mathfrak{A}, \mathfrak{B})$, constructed as follows: one throws a number α into $\mathfrak{A}$ if it is not undercut by any number of the system S, and a number β into $\mathfrak{B}$ if it is undercut by at least one number in S. The number g determined by this cut is not undercut by any number in S, for otherwise there would also be numbers smaller than g, and therefore belonging to $\mathfrak{A}$, which nevertheless are undercut by numbers in S; while, on the other hand, every number β lying even ever so little above g belongs to $\mathfrak{B}$ and is therefore undercut by a number in S. These are precisely the characteristic marks of the lower bound.

2. If S' is a part of S, then the numbers S' also have a lower bound g', and this is either equal to g or greater than g.

For it cannot be smaller than g, since otherwise numbers would occur in S', and therefore also in S, which lie below g.

Now let $f(x)$ be a real function of x which has only finite values in the interval

$$a \le x \le b \tag{1}$$

and which is also continuous in this interval. In agreement with § 31, V and VI, this means that

$$f(x \pm h) - f(x) \tag{2}$$

falls, in absolute value, below an arbitrarily prescribed number η throughout the interval whenever the positive h is smaller than a certain number ε. For $x = a$ we take in (2) only the upper sign, and for $x = b$ only the lower sign, so as not to pass outside the interval with $x \pm h$.

For the values of such a function in the interval there is, by 1., a lower bound g, and we now prove the proposition:

3. The function $f(x)$ assumes the value g for some value ξ of the interval, so that the lower bound becomes a minimum of the function $f(x)$ in the interval.

The proof is as follows. By 2., the lower bound of the function values in a part of the interval (1) is either equal to g or greater than g.

If now $f(a) = g$, what is to be proved is true. If, however, $f(a) > g$, then by the continuity of $f(x)$ one can give an interval $a \le x \le a + h$ in which $f(x) > g$ remains true, and hence the lower bound of $f(x)$ is greater than g.

We now construct in the interval (1) a cut $(\mathfrak{A}, \mathfrak{B})$ of the following kind: we count a value α of the interval (1) as belonging to $\mathfrak{A}$ if the lower bound of $f(x)$ in the interval $a \le x \le \alpha$ is greater than g, and a value β as belonging to $\mathfrak{B}$ if the lower bound in the interval $a \le x \le \beta$ is equal to g.

It is possible that $\mathfrak{B}$ consists only of the single value b; but then $f(b) = g$ must hold, and our assertion is fulfilled for $x = b$. For if $f(b) > g$, one could choose a quantity g' between $f(b)$ and g, and, by continuity, an interval $b - h \le x \le b$ such that in this interval all function values $f(x)$ are greater than g', so that their lower bound is equal to or greater than g', and hence certainly greater than g. But since $b - h$ belongs to $\mathfrak{A}$, the lower bound is also greater than g in the interval $a \le x \le b - h$, and consequently in the whole interval (1), contrary to the assumption.

If therefore $f(b) > g$, the cut $(\mathfrak{A}, \mathfrak{B})$ defines a number ξ in the interior of the interval (1), and we can now show that

$$f(\xi) = g \tag{3}$$

must hold.

If $f(\xi) > g$ and $f(\xi) > g' > g$, then, by the continuity of $f(x)$, we can determine an interval

$$\xi - h \leq x \leq \xi + h$$

in which all function values $f(x)$ are greater than g', and hence their lower bound is greater than g. But since $\xi - h$ belongs to $\mathfrak{A}$, the lower bound of $f(x)$ in the whole interval $a \leq x \leq \xi + h$ is also greater than g, while $\xi + h$ belongs to $\mathfrak{B}$, and this is the contradiction.

Let $f(x)$ now be an integral rational function with arbitrary complex or real coefficients, and let the variable x also be allowed to take complex values.

By § 31, IV, if C is an arbitrary positive value, the positive quantity R can be so chosen that

$$|f(x)| > C \tag{4}$$

whenever

$$|x| \geq R. \tag{5}$$

For visualization we represent the region (R) bounded by the inequality

$$|x| < R$$

for the variable $x = y + iz$ by a circular disc of radius R in the plane of the variable $x = y + iz$.

If we take C larger than any value of $|f(x)|$ in the interior of the region (R), for example larger than $|f(0)|$, then $|f(x)|$ will certainly become smaller in the interior of (R) than on its boundary. Since now in the whole interior of (R) the function $|f(x)|$, which for abbreviation we shall now denote by X, is not negative, there is a lower bound g for the values of X, and we have to prove the proposition:

4. There exists a value ξ of x in the interior of the region (R) such that

$$|f(\xi)| = g, \tag{6}$$

so that here also the lower bound is a minimum.

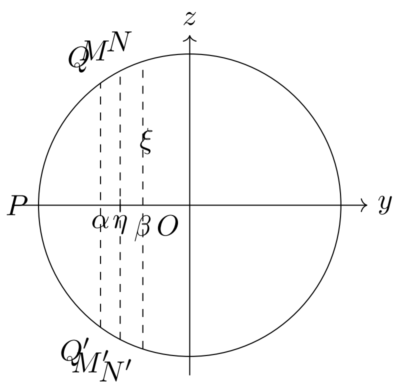
The lower bound of X in any part of the region is either equal to g or greater than g.

A domain of magnitudes determined by the inequalities

$$|x| \leq R, \qquad -R \leq y \leq \alpha \tag{7}$$

is represented in our Figure 2 by the segment $(PQ\alpha Q')$, which we shall call the segment (P, α).

Fig. 2.



We now first determine a value η of y by a cut $(\mathfrak{A}, \mathfrak{B})$ as follows: a number α between $-R$ and $+R$ is admitted into $\mathfrak{A}$ if the lower bound of X in the segment (P, α) is greater than g, and a value β is admitted into $\mathfrak{B}$ if the lower bound of X in the segment (P, β) is equal to g. This cut $(\mathfrak{A}, \mathfrak{B})$ defines a number η with the property that the lower bound of X in the region (P, γ) is greater than g if $\gamma < \eta$, and equal to g if $\gamma > \eta$.

Now, by § 31, VI,

$$|f(\eta + iz)|$$

is a continuous function of z in the interval

$$-\sqrt{R^2 - \eta^2} \leq z \leq \sqrt{R^2 - \eta^2}, \tag{8}$$

which is denoted in Figure 2 by M, M'.

Thus by 3. this function receives, for some value ζ of z in this interval, a minimum value γ, so that, if

$$\xi = \eta + i\zeta$$

is put,

$$|f(\xi)| = \gamma. \tag{9}$$

It is now further easy to see that $\gamma = g$ must be the case; for since g is the lower bound of all values of X within (R), γ cannot first of all be smaller than g.

But γ also cannot be greater than g; for all values which X assumes on the segment MM' are $\geq \gamma$. If, however, $\gamma > g$, then one can choose a quantity g' such that $\gamma > g' > g$, and, by the continuity of X proved in § 31, VI, numbers α, β can be so determined that

$$\alpha < \eta < \beta$$

and such that in the whole region $(P, \beta) - (P, \alpha) = (\alpha, \beta)$ ($QNN'Q'$ in Figure 2) X remains greater than g'. Consequently the lower bound of X in (α, β) is greater than g. Since now the lower bound of X in (P, α) is also greater than g, it is also greater than g in (P, β), which is composed of (P, α) and (α, β); but β belongs to $\mathfrak{B}$, and this would give a contradiction with the definition of $\mathfrak{B}$.

Thus there remains only the possibility that $\gamma = g$, and proposition 4 is thereby proved; namely, there is a value ξ in the interior of (R) for which

$$|f(\xi)| = g$$

holds, so that $|f(x)|$ attains a minimum value. We now prove:

5. If α is any value of x for which $f(\alpha)$ is different from zero, then h can be chosen so that

$$|f(\alpha + h)| < |f(\alpha)| \tag{10}$$

results.

It then follows that, if $f(\xi)$ were different from zero, g could not be the minimum of the function $|f(x)|$; but since this has been proved, one must have

$$f(\xi) = 0, \tag{11}$$

and ξ is a root of the equation $f(x) = 0$. The fundamental theorem will therefore be proved.

In the proof of 5. we make use of the theorem already proved (§ 32), that a pure equation always has a root.

Some of the derived functions $f'(\alpha), f''(\alpha), \ldots$ may vanish; the last, $f^{(n)}(\alpha)$, which is equal to $\Pi(n)a_0$, is however different from zero. Thus suppose that $f'(\alpha), f''(\alpha), \ldots, f^{(m-1)}(\alpha)$ vanish, but $f^{(m)}(\alpha)$ does not vanish. We then have, by § 13,

$$f(\alpha + h) = f(\alpha) + \frac{h^m}{\Pi(m)} f^{(m)}(\alpha) + \frac{h^{m+1}}{\Pi(m+1)} f^{(m+1)}(\alpha) + \cdots. \tag{12}$$

We choose, as is always possible by § 31, V, a positive number ε so that

$$\left|\frac{h}{m+1}\frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)}+\frac{h^2}{(m+1)(m+2)}\frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)}+\cdots\right|<1 \tag{13}$$

whenever

$$|h|<\varepsilon, \tag{14}$$

and a positive proper fraction δ such that

$$\delta|f(\alpha)|<\left|\frac{\varepsilon^m f^{(m)}(\alpha)}{\Pi(m)}\right|, \tag{15}$$

which, if $f(\alpha)$ does not vanish, is likewise possible. We then determine h (on the basis of § 32) from the equation

$$\frac{h^m f^{(m)}(\alpha)}{\Pi(m)}=-\delta f(\alpha), \tag{16}$$

whence, by (15),

$$|h|<\varepsilon$$

follows. From (12), when the value from (16) is substituted for h^m, we now get

$$f(\alpha+h)=(1-\delta)f(\alpha)-\delta f(\alpha)\left(\frac{h}{m+1}\frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)}+\frac{h^2}{(m+1)(m+2)}\frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)}+\cdots\right),$$

and hence

$$\begin{aligned}|f(\alpha+h)|\le{}&|f(\alpha)|(1-\delta)\\&+\delta|f(\alpha)|\left|\frac{h}{m+1}\frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)}+\frac{h^2}{(m+1)(m+2)}\frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)}+\cdots\right|.\end{aligned} \tag{17}$$

By (13), therefore,

$$|f(\alpha+h)|<|f(\alpha)|. \tag{18}$$

Thus the proof of the fundamental theorem is completed.